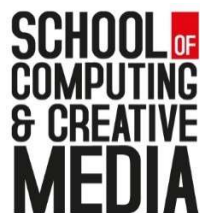


School of Engineering, Computing & Built Environment
School of Computing and Creative Media

Bachelor of Computer Science (Hons)
/ Bachelor of Software Engineering (Hons)

Advanced Programming XBAU2014N

Prepared by Bridget Merliza a/p Paul Christopher Archibald
September 2025 Semester



Task Assignment System

Course Title : ADVANCED PROGRAMMING
Course Code : XBAU2014N
Course Lecturer : Bridget Archibald (bridget.p@uow.edu.my)
: Tan Phit Huan (phithuan.tan@uow.edu.my)

BRIEF

CLO Alignment & Problem Statement

- SLO1: Implement procedural-based programming solution to problems using C++ programming language
- SLO4: Incorporate advanced features in C++ to achieve efficient implementations

The objective of this assignment is to implement a C++ application, Task Assignment System, using procedural programming. The solution should incorporate advanced features in C++ to achieve efficient implementations. To align with modern sustainable computing directives, your application must integrate the UN Sustainable Development Goal (SDG 9: Industry, Innovation and Infrastructure).

REQUIREMENTS :

Part A : Implementation Procedural Programming - Aligned with SLO1

Implement a C++ application to simulate the process of task assignment system. The Tasks.txt and Workers.txt files save the information of individual tasks and workers respectively. Each task represents a task together with a list of candidate workers. We will simulate the trials of all workers in this list, calculate their average performance and choose the best performer. The best performer is the worker with the highest average performance over the trials.

- **Inputs in Tasks.txt and Workers.txt**

- *Tasks.txt*

- *Format:*
taskId,description;uncertainty\$difficulty%priorityLabel&workers:list of worker IDs
 - *Example: 123,image labelling;5\$10%1&workers:0,1*
 - *Note: priorityLabel indicates whether a task has high priority (1 = high, 0 = low).*

- *Workers.txt*

- *Format: workerId,name%variability\$ability;experienceLabel*
 - *Example: 0,Michael%-2\$50;1*
 - *Note: experienceLabel indicates whether a worker is senior (1 = senior, 0 = ordinary).*

- *Worker Performance and Average Performance*

- *Performance score is sampled from a normal distribution where mean = worker ability - task difficulty and standard deviation = |task uncertainty + worker variability|. If the standard deviation is 0, use 1.*
 - *Average performance is the average of multiple independent draws from this distribution plus a conditional value. Use 10 draws for low-priority tasks and 15 draws for high-priority tasks. Add 0 for ordinary workers and 2 for senior workers.*
 - *Summary: Average Performance = Average of {10 or 15} independent performance scores + {0 or 2}.*

Part B : Advanced Modern C++ Requirement - Aligned with SLO4

Students are expected to demonstrate proficiency beyond basic procedural programming constructs by incorporating appropriate modern C++ features into their solution. Marks will be awarded for the effective and meaningful use of features that improve program quality. In addition, all external references used must be cited within the source code and listed in the README file, with a clear indication of how the resources were utilized during development. Unreferenced use of external materials may be treated as academic misconduct.

NOTE

- ✓ Do not modify the provided input files.
- ✓ Use structs to store task and worker information.
- ✓ Vectors are not allowed. Arrays may be used. The number of workers and tasks will not exceed 20.
- ✓ Standard I/O libraries such as stringstream may be used.
- ✓ Include good comments, layout, and concise self-explanatory code.
- ✓ Academic misconduct will result in a zero mark and be reported to the University.

SUBMISSION REQUIREMENTS

- ✓ Create a ZIP file named `campus_studentId_ass1.zip` (e.g. **UGM12321_ass1.zip** or **PGM12322_ass1.zip**) and submit via **MsTeams**.
- ✓ The ZIP file must contain: **driver.cpp**, **header.h**, **header.cpp**, a **README file**, the **Gen AI Declaration**
- ✓ The README must describe the role of each file, how to run the program and reference list.

ASSESSMENT CRITERIA

Award:

50 marks for implementation Procedural Programming

50 marks for Advanced Modern C++ Requirement

DUE DATE:

9th October 2025 (Friday)

Name : _____

(Student id: _____)

LEARNING OUTCOME	Fail (0-49)	3rd Class (50-59)	2nd Lower Class (60-69)	2nd Upper Class (70-79)	1st Class (80-100)	Marks (100%)	Weightage	Actual Marks
SLO1 Procedural Programming Implementation	Program does not meet the assignment requirements. Significant errors in file processing, task assignment logic, calculation of worker performance, or output generation.	Program partially meets requirements with several logic, design, or implementation errors. Limited use of procedural decomposition.	Program correctly implements most requirements with minor errors. Appropriate use of functions, structs, arrays, and file handling.	Program correctly implements all major requirements with good procedural design, modularity, and accurate processing of tasks and workers.	Program fully satisfies all requirements. Demonstrates excellent procedural design, robust validation, accurate calculations, well-structured functions, and high-quality implementation.		0.50	
SLO4 Advanced Modern C++ Features, Documentation & Referencing	Little or no evidence of modern C++ features. README is missing or incomplete. References are absent.	Limited use of modern C++ features with minimal contribution to solution quality. README and references are incomplete.	Appropriate use of some modern C++ features. README adequately describes the solution and includes basic references.	Effective use of several modern C++ features that improve readability, maintainability, and efficiency. README is clear, complete, and references are properly cited in code and README.	Excellent and meaningful integration of modern C++ features. README is comprehensive, professionally presented, and all external sources are consistently cited within the code and README with clear explanations of their use.		0.50	

